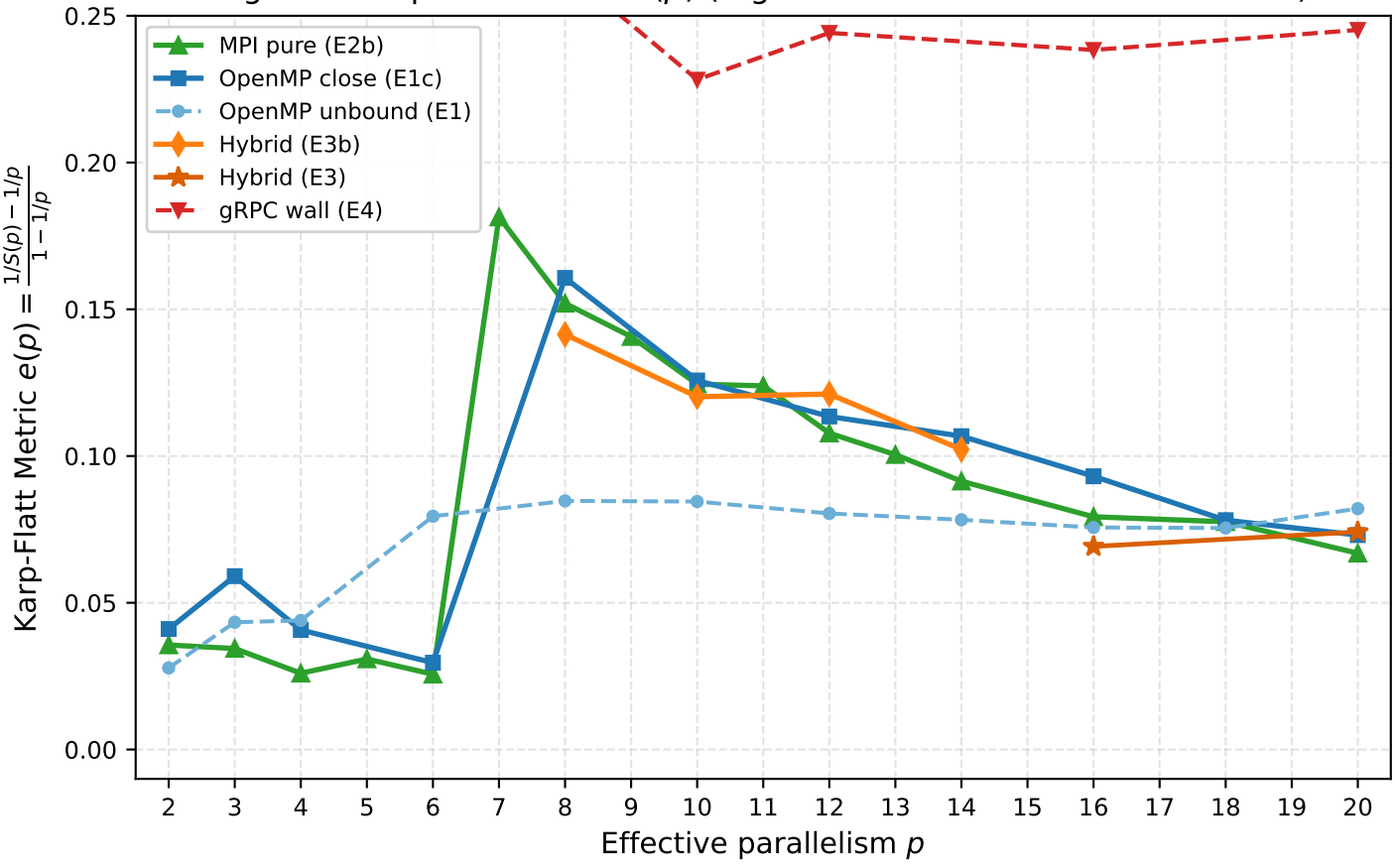
Fig. C — Karp-Flatt Metric $e(p)$ (Algorithmic Overhead vs Contention)
MPI pure (E2b)
OpenMP close (E1c)
OpenMP unbound (E1)
Hybrid (E3b)
Hybrid (E3)
gRPC wall (E4)
Karp-Flatt Metric $e(p) = \frac{1/S(p) - 1/p}{1 - 1/p}$
Effective parallelism p
0.25
0.20
0.15
0.10
0.05
0.00
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20